
Discussion 7

Note: Your TA will probably not cover all the problems on this worksheet. The discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, discussions, and homework.

This Week's Cool AI Demo/Video:

Figure 3 just dropped: <https://www.youtube.com/watch?v=Eu5mYMAvctM>

Sora 2 memes: <https://www.youtube.com/watch?v=7tV1RLQps04>

<https://www.youtube.com/shorts/NCgSr41j1qA>

<https://www.youtube.com/shorts/F1UDA3uzHrQ>

1 Taylor Expansion and Quadratic Forms (shifted minimum)

In Lecture 13, we saw that the local behavior of an error surface $E(\mathbf{w})$ around a stationary point $\mathbf{w}^*$ can be approximated using its second-order Taylor expansion:

$$\tilde{E}_{\mathbf{w}^*}(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2}(\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}(\mathbf{w} - \mathbf{w}^*),$$

where $\mathbf{H} = \nabla^2 E(\mathbf{w}^*)$ is the Hessian matrix describing the local curvature. The eigenvalues and eigenvectors of $\mathbf{H}$ determine the curvature along each principal direction and whether $\mathbf{w}^*$ is a minimum, maximum, or saddle point.

For this question, consider a quadratic error function:

$$E(\mathbf{w}) = \frac{5}{2}w_1^2 - 4w_1w_2 + \frac{5}{2}w_2^2 - w_1 - w_2 + 2.$$

(a) Find the stationary point $\mathbf{w}^*$ of $E(\mathbf{w})$.

Hint: Compute the gradient $\nabla E(\mathbf{w}) = \begin{bmatrix} \frac{\partial E}{\partial w_1} \\ \frac{\partial E}{\partial w_2} \end{bmatrix}$ and set it equal to zero.

Solution:

Compute the partial derivatives:

$$\frac{\partial E}{\partial w_1} = 5w_1 - 4w_2 - 1, \quad \frac{\partial E}{\partial w_2} = -4w_1 + 5w_2 - 1.$$

Setting $\nabla E = \mathbf{0}$ gives the linear system

$$\begin{cases} 5w_1 - 4w_2 - 1 = 0, \\ -4w_1 + 5w_2 - 1 = 0. \end{cases}$$

Solving this linear system, we find

$$w_1 = 1, \quad w_2 = 1.$$

Thus the stationary point is

$$\mathbf{w}^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- (b) Compute the Hessian matrix $\mathbf{H} = \nabla^2 E(\mathbf{w})$ and determine its eigenvalues λ_1, λ_2 and corresponding unit eigenvectors $\mathbf{u}_1, \mathbf{u}_2$. Based on the signs of the eigenvalues, determine whether $\mathbf{w}^*$ is a local minimum, local maximum, or saddle point.

Hint: The Hessian is the matrix of second partial derivatives: $H_{ij} = \frac{\partial^2 E}{\partial w_i \partial w_j}$.

Solution:

The Hessian (second derivatives) is independent of $\mathbf{w}$:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 E}{\partial w_1^2} & \frac{\partial^2 E}{\partial w_1 \partial w_2} \\ \frac{\partial^2 E}{\partial w_2 \partial w_1} & \frac{\partial^2 E}{\partial w_2^2} \end{bmatrix} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}.$$

To find the eigenvalues of H , we compute the roots of $\det(\mathbf{H} - \lambda I) = 0$:

$$\begin{vmatrix} 5 - \lambda & -4 \\ -4 & 5 - \lambda \end{vmatrix} = (5 - \lambda)^2 - 16 = 0 \implies 5 - \lambda = \pm 4.$$

Solving gives us

$$\lambda_1 = 1, \quad \lambda_2 = 9.$$

Using the definition $\mathbf{H}\mathbf{u}_i = \lambda_i \mathbf{u}_i$ and normalizing gives us

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Both eigenvalues are positive, so the stationary point $\mathbf{w}^*$ is a local minimum.

- (c) Use the eigendecomposition of the Hessian to write the 2nd-order Taylor expansion of $E(\mathbf{w})$ around $\mathbf{w}^*$ in the eigenvector basis. Your answer should be in terms of α_1 and α_2 , where α_i is the coordinate of $(\mathbf{w} - \mathbf{w}^*)$ along the direction of $\mathbf{u}_i$.

Solution:

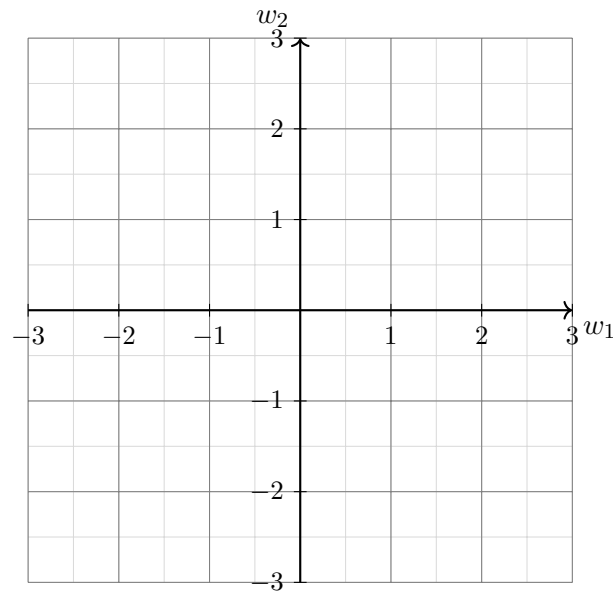
Diagonalize $\mathbf{H} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ with $\mathbf{\Lambda} = \text{diag}(1, 9)$ and $\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2]$. Let $\boldsymbol{\alpha} = \mathbf{U}^\top(\mathbf{w} - \mathbf{w}^*)$ be coordinates in the eigenbasis. Then

$$\tilde{E}_{\mathbf{w}^*}(\mathbf{w}) = E(\mathbf{w}^*) + \frac{1}{2}(\mathbf{w} - \mathbf{w}^*)^\top \mathbf{H}(\mathbf{w} - \mathbf{w}^*) = E(\mathbf{w}^*) + \frac{1}{2}\boldsymbol{\alpha}^\top \mathbf{\Lambda}\boldsymbol{\alpha} = E(\mathbf{w}^*) + \frac{1}{2}(\lambda_1 \alpha_1^2 + \lambda_2 \alpha_2^2).$$

With $E(\mathbf{w}^*) = 1$, $\lambda_1 = 1$, $\lambda_2 = 9$:

$$\tilde{E}(\boldsymbol{\alpha}) = 1 + \frac{1}{2}\alpha_1^2 + \frac{9}{2}\alpha_2^2.$$

- (d) Sketch the isocontours of $E(\mathbf{w})$ and mark $\mathbf{w}^*$. Indicate the eigenvector directions $\mathbf{u}_1$ and $\mathbf{u}_2$, labeling which has the greater curvature.



Solution:

The isocontours of $E(\mathbf{w})$ are ellipses centered at

$$\mathbf{w}^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

From part (b), the eigendecomposition of the Hessian gives:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \lambda_1 = 1; \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \lambda_2 = 9.$$

- The direction of $\mathbf{u}_1$ (along $[1, 1]$) corresponds to the smaller eigenvalue $\lambda_1 = 1$, indicating the **least curvature** and thus the **major axis** of the ellipses.
- The direction of $\mathbf{u}_2$ (along $[1, -1]$) corresponds to the larger eigenvalue $\lambda_2 = 9$, indicating the **greatest curvature** and thus the **minor axis**.

Hence, the ellipses are elongated along the $\mathbf{u}_1$ direction and compressed along $\mathbf{u}_2$, with the center at the minimum $\mathbf{w}^* = (1, 1)$.

2 Momentum vs. SGD

Consider a loss/error function $E(\mathbf{w})$ with parameters $\mathbf{w} \in \mathbb{R}^D$. Recall the stochastic gradient descent (SGD) update rule is

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} - \eta \nabla E(\mathbf{w}^{(\tau)}),$$

where $\eta > 0$ is the learning rate and τ is the iteration number (starting at 0).

We saw in lecture that we can improve upon vanilla SGD by adding momentum, such that

$$\Delta \mathbf{w}^{(\tau-1)} = -\eta \nabla E(\mathbf{w}^{(\tau-1)}) + \boxed{\mu \Delta \mathbf{w}^{(\tau-2)}},$$

$$\mathbf{w}^{(\tau)} = \mathbf{w}^{(\tau-1)} + \Delta \mathbf{w}^{(\tau-1)},$$

where $0 \leq \mu < 1$ is the momentum parameter and $\Delta \mathbf{w}^{(0)} = 0$.

- (a) In a nearly flat region where the gradient direction stays almost constant, how does momentum affect learning compared to plain SGD?
- (i) It slows learning down to prevent overshooting.
 - (ii) It keeps the same pace as SGD.
 - (iii) It speeds learning up by accumulating updates in the same direction.
 - (iv) It randomly perturbs the updates to explore more directions.

Solution:

(iii): In flat regions, the gradient direction barely changes. Each step reinforces the previous one, so momentum “builds speed” along that direction, effectively increasing how far the parameters move each iteration.

Example: Consider a nearly flat loss surface $E(w) = \frac{1}{200}(w - 5)^2$ and start from $w_0 = 0$ with $\eta = 0.1$. Then $\nabla E(w) = \frac{1}{100}(w - 5)$, which changes very little as w moves.

- SGD: $w_1 = 0.005$, $w_2 = 0.0095$, $w_3 = 0.01355, \dots$
- Momentum ($\mu = 0.9$): $w_1 = 0.005$, $w_2 = 0.0145$, $w_3 = 0.0270, \dots$

Momentum quickly accelerates toward $w = 5$ by adding up past gradients.

- (b) In a steep or highly curved region where the gradient direction alternates sign each step (causing oscillations), how does momentum affect the updates?
- (i) It amplifies oscillations by adding more energy to the updates.
 - (ii) It cancels out part of each new step, smoothing and stabilizing motion.
 - (iii) It leaves the oscillations unchanged.
 - (iv) It reverses the update direction completely.

Solution:

(ii): Momentum blends the oscillatory directions over time, causing partial cancellation. This reduces the size of the back-and-forth motion, helping the optimizer move more smoothly.

Example: Consider a narrow valley

$$E(w_1, w_2) = w_1^2 + 10w_2^2,$$

with learning rate $\eta = 0.1$. The gradient is

$$\nabla E = (2w_1, 20w_2),$$

so the slope is gentle in w_1 but steep in w_2 . Starting from $(w_1, w_2) = (1, 1)$:

- **SGD:**

$$w_2^{(1)} = 1 - 0.1(20)(1) = -1.0,$$

$$w_2^{(2)} = -1.0 - 0.1(20)(-1.0) = 1.0,$$

$$w_2^{(3)} = -1.0, \quad \text{and so on.}$$

The w_2 coordinate oscillates between 1 and -1 and will not converge.

- **Momentum ($\mu = 0.9$):**

$$\Delta w_2^{(1)} = -0.1 \cdot 20 \cdot 1 + 0.9 \cdot 0 = -2.0,$$

$$w_2^{(1)} = 1 + \Delta w_2^{(1)} = 1 - 2 = -1.0.$$

$$\Delta w_2^{(2)} = -0.1 \cdot 20 \cdot (-1.0) + 0.9 \cdot (-2.0) = 2.0 - 1.8 = 0.2,$$

$$w_2^{(2)} = -1.0 + 0.2 = -0.8.$$

$$\Delta w_2^{(3)} = -0.1 \cdot 20 \cdot (-0.8) + 0.9 \cdot 0.2 = 1.6 + 0.18 = 1.78,$$

$$w_2^{(3)} = -0.8 + 1.78 = 0.98.$$

Now the w_2 sequence still oscillates between positive and negative values but with damped amplitude, so it will eventually converge (albeit still slowly).